2023_Hexaware_Actual_Test_10

Test Summary

- No. of Sections: 3
- No. of Questions: 92
- Total Duration: 130 min

Section 1 - Aptitude

Section Summary

- No. of Questions: 60
- Duration: 60 min

Additional Instructions:

None

Q1. Find the unit digit for the product $568^{613} \times 2173^{562} \times 29^{29}$.

- 8
- 6
- 9
- 1

Q2. If 5 men with 7 boys can earn Rs.127.50 in 6 days and 2 men with 3 boys can earn Rs.35 in 4 days, then find the time in which 7 men with 6 boys will earn Rs.750.

- 25
- 15
- 20
- 30

Q3. A can do a piece of work in 12 days and B in 16 days. A began the work alone, After some days B joined the work. The work was completed in 9 days, in all. How many days before the completion of the work did B join the work?

- 4
- 12
- 10
- 8

Q4. The average salary of the entire staff in an office is Rs 250 per month. The average salary of officers is Rs 520 and that of non-officers is Rs. 200. If the number of officers is 15, then find the number of non–officers in the office.

- 83
- 81
- 87
- 56

Q5. Five persons went to a hotel for taking their meals. Four of them spent Rs.10 each on their meals and the fifth spent Rs. 4 more than the average expenditure of all the five. What was the total money spent by them?

- 40
- 65
- 50
- 55

Q6. Five letters are sent to different persons and addresses on the five envelopes are written at random. The probability that all the letters do not reach the correct destiny is

- 44/120
- 5/120
- 36/120
- 21/120

Q7. In a class 40% of the students offered Physics 20% offered Chemistry and 5% offered both. If a student is selected at random, find the probability that he has offered Physics or Chemistry only.

- 45%
- 55%
- 36%
- None of these

Q8. Two cars A and B travel from one city to another, at speeds of 72 km/hr and 90 km/hr respectively. If car B takes 1 hour lesser than car A for the journey, then what is the distance (in km) between the two cities?

320

240

360

420

Q9. In covering a certain distance , the speed of A and B are in the ratio of 3:4. A takes 30 minutes more than B to reach the destination.The time taken by A to reach the destination is :

60 min

120 min

90 min

45 min

Q10. A man's income from an investment of Rs.2000 at a certain rate of simple interest together with the income from investing Rs.1500 at 1% higher for 4 years equals Rs.620. the rate of interest at which invested Rs.2000, is _____

4%

5%

7%

$5\frac{1}{2}$%

Q11. Suresh borrows Rs.6375 to be paid back with compound interest at the rate of 4 % pa by the end of 2 year in two equal yearly installments. How much will each installment will be?

3840

3380

4800

Data inadequate

Q12. A, B and C together can finish a task in 12 days. A is twice as productive as B and C alone can do the task in 36 days. In how many days can A and B do the task, if C goes on leave?

18

20

22

12

Q13. A beggar had ten paise, twenty paise, and one rupee coins in the ratio 10: 17: 7 respectively at the end of the day. If that day he earned a total of Rs.57, how many twenty paise coins did he have?

114

171

95

85

Q14. The ratio between the time taken by three cars (A, B, and C) that travel the same distance is 12 : 15 : 10. If V_A = 60 km/hr, find V_B.

20

48

30

25

Q15. A vessel contains 20 liters of a mixture of milk and water in a ratio of 3:2. 10 liters of the mixture are removed and replaced with an equal quantity of pure milk. If the process is repeated once more, find the ratio of milk and water in the final mixture obtained.

9:1

4:7

7:1

2:5

Q16. **Directions:**
What will come in place of the question mark (?) in each of the following number series?

33, 43, 65, 99, 145, (?)

Q17. Direction: What should come in place of question mark (?) in the following number series?
24, 536, 487, 703, 678, ?

Q18. Find the missing term in the series:
7, 12, 19, ?, 39.

- 29
- 28
- 26
- 30

Q19. **Directions:** In the following question, Conclusions are given and five statements are given as a), b), c), d) and e. From this, you have to take the statements to be true even if they seem to be at variance with commonly known facts and then decide which of the given statement logically follows.

Conclusions:
Some rocks are dogs
Some rocks are chains
Some wells are rocks
Some dogs are wells is a possibility

a) Statements: No dogs are chains. All chains are rocks. Some rocks are wells
b) Statements: Some dogs are chains. All chains are rocks. Some rocks are wells
c) Statements: All dogs are chains. All chains are rocks. No rocks are wells
d) Statements: Some dogs are not chains. All chains are rocks. Some rocks are not wells
e) Statements: No dogs are chains. Some chains are rocks. All rocks are wells

- (a)
- (b)
- (c)
- (d)
- (e)

Q20. Directions : Read the conclusions and then decide which of the given conclusions logically follows from the given statements, disregarding commonly known facts.
Give answer:
(a)If only conclusion I follows
(b)If only conclusion II follows
(c)If either conclusion I or II follows
(d)If neither conclusion I nor II follows
(e) If both conclusions I and II follow

Statement:
Only few Books are Pens
Only few Pencils are Pens
No Pencil is Marker
Some Marker is Orange
Conclusions:

I) Some Markers are Pens is a Possibility.
II) Some Books can never be Marker.

- A
- B
- C
- D
- E

Q21. In the question given below is given a statement followed by three conclusions numbered I, II, and III. You have to assume everything in the statement to be true, then consider the three conclusions together and decide which of them logically follows beyond a reasonable doubt from the information given in the statement

Statements: No box is toy. All toys are blocks.
Conclusions:
I. Some blocks are toys.
II. Some blocks are boxes.
III. No block is box.

- Only I follows
- Either II or III follows
- Only either II or III, and I follows
- None follows

Q22. L is the daughter of K. L is married to D. B is the son of D. T is married to B. How is D related to T?

- Father-in-law
- Mother-in-law
- Father
- Mother

Q23. When Martin saw Hannah, he recalled, "She is the daughter of the father of my daughter's mother." Who is Hannah to Martin?

- Sister-in-law
- Sister

Cousin

Aunt

Q24. Pointing to a lady in the photograph, Apoorva said, “Her son’s father is the son-in-law of my mother.” How is Apoorva related to the lady?

Aunt

Sister

Mother

Cousin

Q25. In the question below are given two or three statements followed by two conclusions numbered I and II. You have to take the given statements to be true even if they seem to be at variance from commonly known facts and then decide which of the given conclusions logically follows from the statements, disregarding commonly known facts. Give answer.

Statements :
U ≥ X = Y, Y ≤ Z ≤ S, T = W > Z
Conclusions:
I. T ≤ U
II. S > U

if only conclusion I follows

if only conclusion II follows

if either conclusion I or conclusion II follows

if neither conclusion I nor conclusion II follows

if both conclusions I and II follows

Q26. Direction: In the following question, the symbols θ, ɲ, &, %, $\$$ are used with the following meaning as illustrated below:

‘A θ B’ means ‘A is not smaller than B’.
‘A ɲ B’ means ‘A is neither smaller than nor equal to B’.
‘A & B’ means ‘A is neither greater than nor equal to B’.
‘A % B’ means ‘A is neither greater than nor smaller than B.
‘A $\$$ B’ means ‘A is not greater than B’.

Now in the following question assuming the given statements to be True, find which of the conclusion/s given below them is/are definitely True?

Statements:
D % Q, Q ɲ L, L θ B, B & G

Conclusions:
I. D ɲ B
II. B % D
III. G ɲ L

Either Conclusion II or III is True.

Both Conclusions I and II are True.

Only Conclusion I is True.

Only Conclusion II is True.

None of these

Q27. The question below is followed by three statements numbered I, II and III. You have to decide whether the data provided in the statements are sufficient to answer the question.

Two items A and B were sold at the same percentage of profit and loss respectively. The loss suffered on item B was Rs. 1683. By what percentage was item B marked up?

I. The original cost price of item A was Rs. 4400.
II. The total loss incurred on both A and B together was Rs. 231
III. Both the items together were marked up by Rs. 808.

If the data in statements II and either I or III together are sufficient to answer the question.

If the data in statements III and either I or II together are sufficient to answer the question.

If the data in all three statements together are not sufficient to answer the question

If the data in statements I and either II or III together are sufficient to answer the question

If the data in all the three statements together are necessary to answer the question

Q28. The question below consists of a question and two statements numbered I and II given below it. You have to decide whether the data provided in the statements are sufficient to answer the questions. Read both the statements and give an answer.

If A and B are positive integers and B is a multiple of A, is A = 2?
(A) $A \neq 1$
(B) B + 2 is a multiple of A.

if statement (A) alone is sufficient to solve the question, but statement (B) alone is not.

if statement (B) alone is sufficient to solve the question, but statement (A) alone is not.

if neither statement (A) nor statement (B) is individually sufficient to solve the question, but a combination of both is sufficient to solve the question

if both the statements taken together are not sufficient and more information is required to solve the question.

Q29. **In the question below relationship between different elements is shown in the statements. The statements are followed by two conclusions**.

Statement: $X > P \geq W < Z \leq T < S, P \geq V$

Conclusions:
I. $P > T$
II. $X > V$

if only conclusion 2 is true

if only conclusion 1 is true

if either conclusion 1 or conclusion 2 is true

if neither conclusion 1 nor conclusion 2 is true

if both conclusion 1 and conclusion 2 are true

Q30. Shabnam travels 9 km towards the west and then 6km towards her left. Then, she turns towards the east and travels 2 km. Finally, she travels 5 km towards the north. How far is she from the starting point?

$5\sqrt{3}$ km

5 km

$5\sqrt{2}$ km

None of these

Q31. **Pick out the best alternative for the underlined part of the sentence.**

In all the big cities, there is a flood of health clubs which claim to build stamina, decrease stress level and weight is reduced.

decrease stress level and weight is reduced.

decrease stress level and weight reduces

decrease stress level and reduce weight

decrease stress level and reduces weight

Q32. Directions: In this question, a sentence has been given with some of its part in bold. To make the sentence grammatically and contextually correct you have to replace the bold part with the correct alternative given below. If the sentence is correct as it is, mark 'No correction required' as your answer.

In some regions around South Asia, French luxury brands **have succeeded in getting customers who might have once thought the cosmetics were out of their league in buying and using for some snazzy getups**.

- have succeeded in getting people who might have once thought the cosmetics were out of their league in buying and using for some snazzy getups
- have succeeded in getting people who might have once thought the cosmetics were out of their league to buy and lease for some snazzy getups
- have succeeded in getting people who once might have thought the cosmetics were out of their league to buy and use for some snazzy getups
- has succeeded in getting people who might have once thought the cosmetics were out of their league to buy and use for some snazzy getups
- has succeeded in getting people who might have once thought the cosmetics were out of their league in buying and using some snazzy getups

Q33. Select the word which is closest in meaning to the given word.
Transient

- perpetual
- changeable
- different
- short-lived

Q34. Choose the word which conveys the **OPPOSITE** meaning for the word:

Farcical

- Serious
- Absurd
- Ludicrous
- Facetious

Q35. Select the most suitable synonym

agnostic

- theist

dogmatist

conformist

skeptic

Q36. **From the given options choose the one which best expresses the given sentence in active/passive voice:**

Have you finished the test?

Has the test finished by you?

Has the test been finished by you?

Had the test been finished by you?

Did you finish the test?

Q37. **From the given options choose the one which best expresses the given sentence in active/passive voice:**

She has learned her lessons.

Her lessons has learned by her.

Her lessons have been learned by her.

Her lessons had been learned by her.

Lessons are learnt by her

Q38. Fill in the blank with a suitable article from the options given.

Gold is …….. precious metal.

a

an

the

no article required

Q39. **Fill in the blank with right preposition:**

He had to repent _____ what he had done.

at

of

over

for

Q40. **Replace the underlined part of the sentence with a grammatically correct part**

While all states face similar industrial waste problems, the predominating industries and the regulatory environment of the states obviously determines the types and amounts of waste produced, as well as the cost of disposal.

all states face similar industrial waste problems, the predominant industries and the regulatory environment of each state obviously determine

each state faces a similar industrial waste problem, their predominant industries and regulatory environment obviously determine

all states face a similar industrial waste problem; their predominating industries and regulatory environment obviously determines

each state faces similar industrial waste problems, the predominant industries and the regulatory environment of each state obviously determines

Q41. Rs.200,000 was invested by Mahesh in an FD @ 10% pa at CI. However every year he has to pay 20% tax on the CI. How much money does Mahesh have after 3 years?

215662.4

216662.4

217662.4

None of these

Q42. The rate of interest for the first 3 years is 6% per annum, for the next 4 years, 7% per annum and for the period beyond 7 years, 7.5% per annum. If a man lent out Rs.1200 for 11 years, find the total interest earned by him.

Rs.1002

Rs.912

Rs.864

Rs.948

Q43. In a chessboard, there are a number of tiles (squares) present in it which is painted only using two different colors (mostly black and white) at alternate places. How many squares are there in the chess board, if we do not consider the 1*1 tile?

- 64
- 140
- 224
- 204

Q44. The manager of a football team of 11 players wants to take photographs of his team, six players at a time. The manager can choose any one out of four renowned photographers. In how many ways can the photographs be taken if the captain and the vice-captain of the team are always included in the photographs?

- ${}^{9}C_{4} \times 6! \times 4$
- ${}^{4}P_{9} \times 9! \times 2$
- ${}^{9}C_{4} \times 9$
- None of these

Q45. How many different integers can be expressed as the sum of three distinct numbers from the set {3,8,13,18,23,28,33,38,43,48}

- 421
- 20
- 22
- 25

Common Content:

Study the information given below and answer the questions based on it.
Seven friends named–P, R, M, L, D, S and N live on seven different floors of a building, but not necessarily in the same order. The lowermost floor of the building is numbered one the above that is numbered two and so on till the topmost floor is numbered seven. Each one of them likes different companies Laptop viz, Samsung, Dell, Lenovo, HP, Acer, Apple and Sony but not necessarily in the same order. Only three people live between R and D. R lives above D. D doesn't live on the lowermost floor. Only one people live between R and the one who likes Lenovo. The one who likes Sony live on an even numbered floor above the one who likes Lenovo. Only two people live between N and the one who likes Lenovo. The one who like samsung lives immediately above N. P lives immediately above S. P does not like Lenovo. The one who likes HP lives on an odd numbered floor below S. N does not like HP. L live above M. Only one person lives between the one who likes Acer and the one who likes Apple. L does not like Acer. N does not like Dell.

Q46. Which of the following combinations is true with respect to the given arrangement?

- Sony - M

HP - R

Acer - D

Lenovo - S

Q47. R likes which of the following laptops?

Samsung

HP

Sony

Dell

Q48. Who amongst the following lives on the floor numbered 4?

The one who likes Apple

The one who likes Lenovo

P

S

Q49. Which of the following statements is true with respect to the given arrangement?

Only two people live between D and N

The one who likes Apple lives immediately above D

M likes Lenovo

M lives on an even numbered floor

Q50. If all the people are made to sit in alphabetical order from top to bottom, the positions of how many people will remain unchanged?

None

One

Three

Two

Common Content:

Read the passage and answer the question given below.

In choosing a method for determining climatic condition that existed in the past, paleo-climatologists invoke four principal criteria. First, the material-rocks, lakes, vegetation, etc on which the method relies must be widespread enough to provide plenty of information, since analysis of material that is rarely encountered will not permit correlation with other regions or with other periods of geological history. Second, in the process of formation, the material must have received an environmental signal .that reflects a change in climate and that can be deciphered by modern physical or chemical means. Third, at least some of the material must have retained the signal unaffected by subsequent changes in the environment. Fourth, it must be possible to determine the time at which the inferred climatic conditions held. This last criterion is more easily met in dating marine sediments, because dating of only a small number of layers in a marine sequence allows the age of other layers to be estimated fairly reliably by extrapolation and interpolation. By contrast, because sedimentation is much less continuous in continental regions, estimating the age of a continental bed from the known ages of beds above and below is more risky.

One very old method used in the investigation of past climatic conditions involves the measurement of water levels in ancient lakes. In temperate regions, there are enough lakes for correlations between them to give us a reliable picture. In arid and semiarid regions, on the other hand, the small number of lakes and the great distances between them reduce the possibilities for correlation. Moreover, since lake levels are controlled by rates of evaporation as well as by precipitation, the interpretation of such levels is ambiguous. For instance, the fact that lake levels in the semiarid southwestern United States appear to have been higher during the last ice age than they are now was at one time attributed to increased precipitation. On the basis of snow-line elevations, however, it has been concluded that the climate then was not necessarily wetter than it is now, but rather that both summers and winters were cooler, resulting in reduced evaporation.

Another problematic method is to reconstruct former climates on the basis of pollen profiles. The type of vegetation in a specific region is determined by identifying and counting the various pollen grains found there.

Although the relationship between vegetation and climate is not as direct as the relationship between climate and lake levels, the method often work well in the temperature zones. In arid and semiarid regions in which is not much vegetation however, small changes one or a few plant types, can change the picture dramatically, making accurate correlations between neighbouring areas difficult to obtain.

Q51. It can be inferred from the passage that an environmental signal found in geological material would not be useful to paleo-climatoiogists if it _____.

- Reflected a change in climate rather than a long-term climatic condition.
- Was incorporated into a material as the material was forming
- Also reflected subsequent environmental changes.
- Was contained in a continental rather than a marine sequence

Q52. According to the passage, the material used to determine past climatic conditions must be widespread for which of the following reasons?
(1) Paleoclimatologists need to make comparisions between periods of geological history.
(2) Paleoclimatologists need to compare materials that have supported a wide variety of vegetation.
(3) Paleoclimatologists need to make comparisons with data collected in other regions.

- 1 only
- 2 and 3 only
- 2 only
- 1 and 3 only

Q53. Which of the following statements best describes the organization of the passage as a whole?

- The author describes a method for determining past climatic conditions and then offers specific examples of situations in which it has been used.

The author discusses the method of dating marine and continental sequences and then explains how dating is more difficult with lake levels than with pollen profiles.

The author describes the common requirements of methods for determining past climatic conditions and then discusses examples of such methods.

The methods for determining past climatic conditions and then describes two of the earliest known methods.

Q54. Which of the following statements about the difference between marine and continental sedimentation is supported by information in the passage?

Data provided by dating marine sedimentation is more consistent with researchers' findings in other disciplines than is data provided by dating continental sedimentation.

It is easier to estimate the age of a layer in a sequence of continental sedimentation than it is to estimate the age of a layer in a sequence of marine sedimentation.

Researchers are more often forced to rely on extrapolation when dating a layer of marine sedimentation than when dating a layer of continental sedimentation.

Marine sedimentation is much more continuous than is continental sedimentation.

Q55. It can be inferred from the passage that paleo-climatologists have concluded which of the following on the basis of their study of snow-line elevations in the southwestern United States?

There is usually more precipitation during an ice age because of increased amounts of evaporation.

Lake levels in the semiarid southwestern United States were lower during the last ice age than they are today.

During the last ice age, cooler weather led to lower lake levels than paleo-climatologists had previously assumed.

The high lake levels during the last ice age may have been a result of less evaporation rather than more precipitation.

Common Content:

Indian cities offer new _____1_____ for development across various urban sectors, but most of these cities ____2_____ from urban flooding during monsoon. This eventually leads to loss of life and property and ____3_____the economic activities and natural eco-system.Even the most planned Indian cities suffer from this problem. For instance, when Chandigarh, which is ___4_____ known as the first planned city of independent India, receives high rainfall, _____5_____ find it difficult to reach their destinations.

Q56. Choose the right word for the blank numbered (4)

angrily

perfectly

famously

proudly

Q57. Choose the right word for the blank numbered (1)

adversity

opportunities

threat

dedication

Q58. Choose the right word for the blank numbered (5)

transporters

commuters

canard

guise

Q59. Choose the right word for the blank numbered (3)

affects

effects

agitate

consequences

Q60. Choose the right word for the blank numbered (2)

suffer

agonize

endure

tolerate

Section 2 - Pseudocode and Computer Fundamentals

Section Summary

- No. of Questions: 30
- Duration: 30 min

Additional Instructions:

None

Q1. What will be the output of the following pseudocode?

```
1  Integer a,b,c
2  Set a=0,b=4,c=8
3  if ((c+a+b)>(b-c))
4     b=(c+c)+a
5     c=(c+c)+a
6  End if
7  if ((c^a^b)<(b^c))
8     a= 3 & a
9  End if
10 Print a+b+c
```

- 32
- 38
- 33
- 29

Q2. What will be the output of the following pseudocode?

```
1  Integer p,q,r
2  Set p=4, q=2, r=12
3  for(each r from 4 to 6)
4    p= (r+7)+p
5    q= (q&11)+p
6  End for
7  for (each r from 2 to 4)
8    q= (4+9)+r
9  End for
10 Print p+q
```

- 56
- 62
- 57
- 60

Q3. What will be the output of the following pseudocode?

```
1  Integer j, m
2  Set m = 2
3  Integer a[4] = { 2,1,1,2 }
4  m = m^a[2]
5  if (a[1]>1)
6    m = m^1
7  Else
8    m = m^2
9  End if
10 Print m
```

8

12

1

-1

Q4. What will be the output of the following pseudocode for x = 23 and y = 222?

```
1  Integer fun1 (Integer x, Integer y)
2    while (y NOT equals 0)
3      Integer c = x & y
4      x = x ^ y
5      y = c << 1
6    end while
7    return x
8  End function fun1()
```

245

22

2223

1022

Q5. What will be the output of the following pseudocode?

```
1  Integer p, q, r
2  Set p=2, q=6, r=5
3  q=q+p
4  r=p+p
5  p=r+q
6  p=(r+q)&q
7  Print p+q+r
```

20

8

30

24

Q6. What will be the output of the following pseudocode?

```
1  Integer p, q, r
2  Set p = 0, q = 3, r = 14
3  if (1<p && (q+p) >(p+q))
4     r = q + r
5  End if
6  print p + q + r
```

17

19

28

14

Q7. What will be the output of the following pseudocode?

```
1  Integer p,q,r
2  Set p=5, q=8, r=8
3  for (each r from 4 to 5)
4   if((1+9)<(10-p))
5     p=p+q
6   End if
7   p= (p+r)+r
8   q=(12&11)+p
9  End for
10 Print p+q
```

51

54

62

56

Q8. What will be the output of the following pseudocode for **x = 343**?

```
1  Integer fun (Integer x)
2     if ((x MOD 10) EQUALS (x / 10))
3        return (x / 7)
4     else
5        return fun (fun (x/7))
6     end if
7  end function fun()
```

3

1

4

0

Q9. What will be the output of the following pseudocode?

```
1  Integer j,m
2  Set m=1
```

```
Integer a[4]={9,1,3,4}
if(a[0]>a[1])
   m=m+a[3]+a[2]-a[1]-a[0]
End if
Print m
```

- 5
- 9
- -3
- -2

Q10. What will be the output of the following pseudocode?

```
Integer a,b,c,d
Set a=7,b=1,c=2
while ((a*a)>(2*a*b)
   d=a+b+12
   a=a+2
   b=b+2
end while
Print d
```

- 23
- 20
- 25
- 28

Q11. What will be the output of the following pseudocode for a=6, b=4, and c=9?

```
Integer funn(Integer a, Integer b,Integer C)
   if (a>c || (9^5)<a)
      a=2+a
      c=b+a
   End if
return a+b+c
```

25

19

14

30

Q12. What will be the output of the following pseudocode?

```
Integer a[],k,t,m
Set a[]={25,20,30,18,17}
Set t=0
for (each k from 0 to 4)
   t=t+a[k]
   if(t mod 2 EQUALS 1)
      Print true
   otherwise
      Print false
   end if
end for
m=t/5
Print m
```

true true true true false
22

false true true true false
20

true false true true false
15

true false true true false
20

Q13. What will be the output of the following pseudocode for a=8,b=3, and c=5?

```
Integer funn(Integre a, Integer b,Integer c)
   a=(b+c)&c
   a=(a+6)+c
   a=2+a
   a=(3+5)+a
   b=(b+8)+a
   return a+b+c
end function funn()
```

75

44

58

64

```
Integer x, y, I, z
Set x = 11 , y = 10
```

```
3  y = y - 5
4  z = y = 3
5  I = x AND z
6  Print z, I
```

- 4 0
- 3 3
- 6 1
- 2 0

Q15. What will be the output of the following pseudocode?

```
1  Integer p,q,r
2  Set p=2, q=8, r=7
3  p=(q+1)+q
4  if((p^q^r)<(q+r+p))
5    r=(12+12)^p
6    p=(p&p)+r
7  End if
8  Print p+q+r
```

- 53
- 43
- 41
- 49

Q16. What is the full form of UPS?

- United Parcel Service
- Uniform Product Supply
- Under Power Supply
- Uninterrupted Power Suplly

Q17. The CPU consists of

- Input, output and processing

- Control unit, primary storage and secondary storage
- Control unit, arithmetic logic unit, and primary storage
- Input, processing, and storage

Q18. General-purpose computers are those that can be adapted to countless uses simply by changing its _____.

- output device
- input device
- processor
- program

Q19. People who gain unauthorized access to computers for the purpose of doing damage are called _____.

- Employees
- Hackers
- Crackers
- Members of organized crime

Q20. Choose the false one:

- Diffie-Hellman uses the same key for encryption & decryption.
- In RSA, both key are private
- DES is a block cipher algorithm
- Vigenere Cipher is poly-alphabetic in nature

Q21. The transport layer protocol used for real-time multimedia, FTS, DNS, and email respectively are

- TCP, UDP, UDP and TCP
- UDP, TCP, TCP and UDP
- UDP, TCP, UDP and TCP

TCP, UDP, TCP and UDP

Q22. What is a network topology?

The physical layout of network devices

The software used for networking

The speed of the network

The type of data transmitted

Q23. Which routing protocol uses a distance-vector algorithm?

OSPF (Open Shortest Path First)

RIP (Routing Information Protocol)

BGP (Border Gateway Protocol)

EIGRP (Enhanced Interior Gateway Routing Protocol)

Q24. What is the primary purpose of the Data Link Layer in the OSI model?

End-to-End Communication

Physical Transmission

Error Detection

Link Establishment and Management

Q25. Which error control policy uses acknowledgment with a sliding window for efficient data transmission?

Stop-and-Wait

Go-Back-N

Selective Repeat

Parity Checking

Q26. What issue does character count framing method face if there is an error during transmission?

Frame loss

Frame misalignment

Destination or receiver synchronization loss

Excessive overhead

Q27. If IRTT = 45 sec, NRTT = 60 sec, α = 0.9, and the Initial deviation is 8 sec, then the Time out is _____.

80.5

81.3

82.5

80.0

Q28. Calculate the effective throughput for transferring a 1000 KB file assuming TCP using a slow start congestion control technique. Given the round trip time of 100 ms, and the maximum segment size is 1460 bytes. Assume there are no losses and both the bandwidth and the receiver window size are infinite.

5 MBPS

10 MBPS

1 MBPS

2 MBPS

Q29. A 40 Mbps broadcast network that controls medium access using polling has 20 hosts and time required for polling the next host is 80 μsec whenever a node is polled, it is allowed to transmit 4000 bytes. Find the efficiency of the broadcast channel

100/9

100/11

80/7

10/11

Q30. If the initial sequence number is 1 and it is incremented the counter by 2,56,000 for every 2 sec, how long does it take for the counter to wrap around?

33,554 seconds

44,554 seconds

33,555 seconds

44,455 seconds

Section 3 - Coding

Section Summary

- No. of Questions: 2
- Duration: 40 min

Additional Instructions:

None

Q1. **Problem Statement**

Given a string **s** and a dictionary of strings **wordDict**, return true if s can be segmented into a space-separated sequence of one or more dictionary words.

Note that the same word in the dictionary may be reused multiple times in the segmentation.

Example 1:
Input: s = "applepenapple", wordDict = ["apple","pen"]
Output: true
Explanation: Return true because "applepenapple" can be segmented as "apple pen apple". Note that you are allowed to reuse a dictionary word.

Example 2:
Input: s = "catsandog", wordDict = ["cats", "dog", "sand", "and", "cat"]
Output: false

Input Format

The first line of input consists of a string **s**, representing the input string.
The second line consists of an integer **n**, indicating the number of words in the dictionary.
The next line consists of **n** space-separated strings representing the words of the dictionary.

Output Format

If the input string can be segmented into words from the dictionary, output "true".
If the input string cannot be segmented into words from the dictionary, output "false".

Refer to the sample output for the formatting specifications.

Constraints

1 ≤ s.length ≤ 100
1 ≤ wordDict.length ≤ 10
1 ≤ wordDict[i].length ≤ 100
s and wordDict[i] consist of only lowercase English letters.
All the strings of wordDict are unique.

Sample Input

```
applepenapple
2
apple pen
```

Sample Output

```
true
```

Sample Input

```
catsandog
5
```

Sample Output

```
false
```

```
cats dog sand and cat
```

Time Limit: - ms Memory Limit: - kb Code Size: - kb

Q2. **Problem Statement**

Given an array nums with n integers, your task is to check if it could become non-decreasing by modifying at most one element.

We define an array as non-decreasing if nums[i] <= nums[i + 1] holds for every i (0-based) such that (0 <= i <= n - 2).

Example 1:
Input:
3
4 2 3
Output:
4 2 3
true
Explanation: You could modify the first 4 to 1 to get a non-decreasing array.

Example 2:
Input:
3
4 2 1
Output:
4 2 1
false
Explanation: You cannot get a non-decreasing array by modifying at most one element.

Input Format

The first line of input consists of an integer n, representing the size of the array numbers.
The second line of input consists of n integers separated by space arr[i], representing the elements of the array numbers.

Output Format

The first output displays the space-separated integers, representing the input array.
The second output displays a single line containing "true" if it is possible to make the array non-decreasing by modifying at most one element, and "false" otherwise.

Refer to the sample output for the formatting specifications.

Constraints

$1 \le n \le 10$
$1 \le arr[i] \le 10^9$

Sample Input

```
3
4 2 3
```

Sample Output

```
4 2 3
true
```

Sample Input

```
3
4 2 1
```

Sample Output

```
4 2 1
false
```

Time Limit: - ms Memory Limit: - kb Code Size: - kb

Answer Key & Solution

Section 1 - Aptitude

Q1

8

Solution

No Solution

Q2

30

Solution

5 men + 7 boys earn Rs.127.50 in 6 days.

5 men + 7 boys earn Rs.21.25 in 1 day .... (i)

2 men + 3 boys earn Rs.35 in 4 days.

2 men + 3 boys earn Rs.8.75 in 1 day .... (ii)

Multiply (i) and (ii) by 2 and 5 respectively so that the relations may become the same.

10 men + 14 boys earn Rs.42.50 in 1 day ..... (iii)

10 men + 15 boys earn Rs.43.75 in 1 day .... (iv)

Subtracting (iii) from (iv), we get 1 boy who earns Rs.1.25 in 1 day.

From (i), 5 men earn Rs.12.50 in 1 day.

=> 1 man earns Rs.2.50 in 1 day.

=> 7 men + 6 boys earn Rs.25 in 1 day.

=> 7 men + 6 boys earn Rs.750 in 30 days.

Q3

4

Solution

A finishes the work in 12 days,

Part of work done by A in 1 day = 1/12

B finishes the work in 16 days,

Part of work done by B in 1 day = 1/16

Let's assume that B joined the work x day before the completion,

∴ B worked for x days, and A worked for 9 days.

9/12+x/16=1

⇒(3/4)+(x/16)=1

⇒ 12 + x = 16

⇒x = 4

Hence, B joined 4 days before the completion of the work.

Q4

81

Solution

Let the required number of non–officers = x

200x + 520 x 15 = 250 (15 + x)

250x – 200x = 520 * 15 – 250 x 15

50x = 4050

x = 81

Q5

55

Solution

Let the average expenditure of all the five be Rs.x

Then, (10 * 4) + (x + 4) = 5x

40 + x + 4 = 5x

4x = 44

x = 11

Total money spent = 5x = (5 * 11) = Rs.55

Q6

44/120

Solution

∵ Probability = [No of Favorable Outcomes/No. of Total Outcomes]

No. of Total Outcomes = Total No. of ways in which 5 envelopes can be sent to 5 persons = 5! = 120

All the letters do not reach the correct destiny can be found using Dearrangement.

D(N)= 5![1/2!-1/3!+1/4!-1/5!] = 120(1/2 - 1/6 + 1/24 - 1/120)= 120(44/120)= 44

Probability = 44/120

Q7

55%

Solution

n(S) = 100

n(A) = 40,

n (B) = 20,

n (A ∩ B) = 5.

P (A ∪ B) = P (A) + P (B) – P (A ∩ B)

40/100 + 20/100 – 5/100

P(A ∪ B) = 55/100 = 55%.

Q8

360

Solution

Let the time taken by A be x hrs, therefore time taken by B will be (x – 1) hrs

From the problem statement

⇒ 72x = 90(x – 1)

⇒ x = 5 hrs

Distance between cities = 72 × 5 = 360 kms

∴ Distance between A and B is 360 kms.

Q9

120 min

Solution

Ratio of speed =3:4

Ratio of time = 4 : 3

Let A takes 4x hrs , B takes 3x hrs

Then 4x - 3x = 30/60

x = 1/2 hr

Time taken by A to reach the destination is 4x = 4 * (1/2) = 2 hr

Q10

4%

Solution

Let the rate of interest at which Rs.2000 were invested be x% per annum.

Rate % for Rs.1500 =(x + 1) %

2000 * x * 4/100 + (x+1)* 1500 * 4/100 =620

Or, 80x + 60x + 60 =620

140x =560

X =560/140

X =4%

Q11

3380

Solution

25x/26 + 625/676x = 6375

x = (6375 * 676)/1275 = 3380

Q12

18

Solution

Let A finish the task in ‘x’ days

Then B can finish the task in ‘2x’ days

∵ C can finish the task in 36 days

A, B and C together can finish a task in 12 days

1/x + 1/2x + 1/36 = 1/12

⇒ x = 27

A and B can finish the task in 27 days and 54 days respectively.

∴ A and B together can finished the task = (54 × 27)/(54 + 27) = 18 days

Q13

85

Solution

10P 20P 100P

10 : 17 :7

Total amount is (100x + 340x + 700x) =5700

1140x =5700

X=5

Total number of 20 p coins

17 x 5 =85

Q14

48

Solution

Let the time taken by car A be 12t and car B be 15t.

Since they cover the same distance V_A*12t =V_B*15t

=> V_B = 60*12t/15t

= 48 km/hr

Q15

9:1

Solution

Milk = 3/5 * 20 = 12 liters, water = 8 liters

If 10 liters of mixture are removed, amount of milk removed = 6 liters and amount of water removed = 4 liters.

Remaining milk = 12 - 6 = 6 liters

Remaining water = 8 - 4 = 4 liters

10 liters of pure milk are added, therefore total milk = (6 + 10) = 16 liters.

The ratio of milk and water in the new mixture = 16:4 = 4:1

If the process is repeated one more time and 10 liters of the mixture are removed, then amount of milk removed = 4/5 * 10 = 8 liters.

Amount of water removed = 2 liters.

Remaining milk = (16 - 8) = 8 liters.

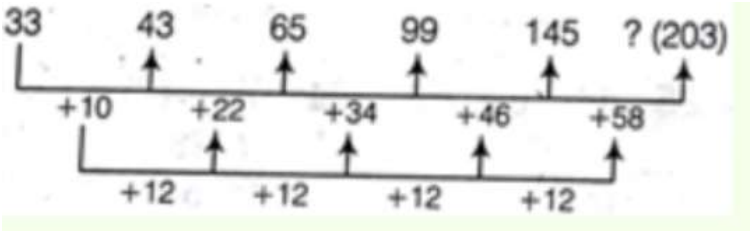
Remaining water = (4 -2) = 2 liters.

The required ratio of milk and water in the final mixture obtained = (8 + 10):2 = 18:2 = 9:1.

Q16

203

Solution



Q17 742

Solution

Q18 28

Solution

Clearly, the given sequence follows the pattern :

+ 5, +7, +9 ... i.e.,

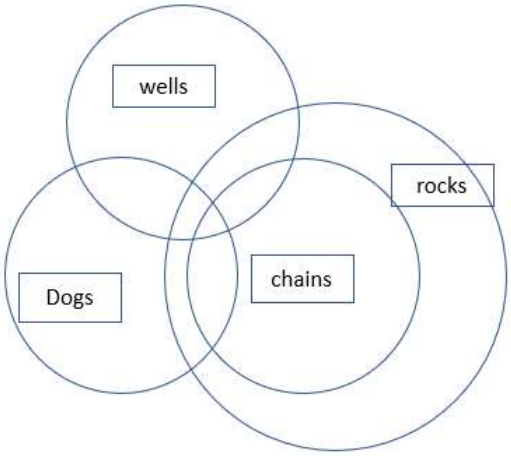
7 + 5 = 12,

12 + 7=19

Missing number = 19 + 9 = 28.

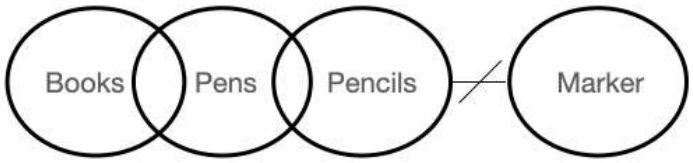
Q19 (b)

Solution



Q20 A

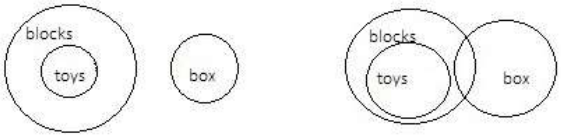
Solution



Q21

Only either II or III, and I follows

Solution



Only either II or III ,and I follows

Q22

Father-in-law

Solution

No Solution

Q23

Sister-in-law

Solution

Martin’s daughter’s mother — Martin’s wife;

Martin’s wife’s father — Martin's father-in-law;

Father-in-law’s daughter — Martin’s sister-in-law or wife.

So, Hannah is Martin’s sister-in-law.

Q24

Sister

Solution

Lady’s son's father is lady’s husband. So, the lady’s husband is the son-in-law of Apoorva’s mother, i.e. the lady is the daughter of Apoorva’s mother. Thus, Apoorva is the lady’s sister.

Q25

if neither conclusion I nor conclusion II follows

Solution

if neither conclusion I nor conclusion II follows

Q26

Only Conclusion I is True.

Solution

No Solution

Q27

If the data in all three statements together are not sufficient to answer the question

Solution

From I and II together

Loss on B = 1683

Total loss = 231

So, profit on A = 1683 - 231 = 1452

So, A was sold at a profit of (1452/4400) * 100% = 33%

So, CP of B = 1683/0.33 = 5100

But we cannot determine the marked up price of B. So, I and II are not sufficient together.

From II and III together

Let Cp of A and B be Rs. a and Rs. b respectively.

So, Profit on A = 1683 - 231 = 1452

As the percentages of profit and loss are equal, so

1452/a = 1683/b

Which is still insufficient to find CP of B. Hence II and III are not sufficient toegther.

From I and III together :

CP of A = 4400

Let CP of B be Rs. B

So, SP of B = b - 1683

Again CP of B cannot be found. So, I and III are not sufficient together.

From I, II and III together :

From I and II, we have CP of B = 5100

CP of A = 4400

Now, total mark up = 808

We cannot determine by how much each item was marked up by.

So, all the statements together are not sufficient.

Q28 if neither statement (A) nor statement (B) is individually sufficient to solve the question, but a combination of both is sufficient to solve the question

Solution

Since B is a multiple of A, there is an integer n such that B = nA.

Statement (A) is clearly not sufficient.

For example, B = 4, A = 2 and B = 9, A = 3 both satisfy Misinformation given in-the question.

From statement (B) it follows that B + 2 = mA for some integer . Since B = nA, nA + 2 = mA or 2 = (m - n)A. Since A is positive, it follows that A = 1 or 2. Note that two values for A are determined and not a unique value.

Thus, statement (B) is not sufficient and the answer must be 3 or 4.

From (A) and (B) together it can be concluded that A = 2.

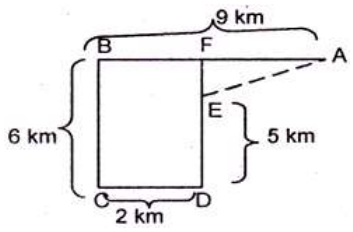
Q29 if only conclusion 2 is true

Solution

X > P ≥ W < Z ≤ T < S No realtionship can be established between P and T. Hence conclusion I is not true. X > P ≥ V X > V. Hence conclusion Ii is true.

Q30 $5\sqrt{2}$ km

Solution



From the figure,

FD = BC = 6 km,AB = 9 km;ED = 5 km, FB = CD = 2 km

∴ EF = BC − ED

= 6 − 5 = 1 km

FA = AB − FB

= (9 − 2) km = 7 km.

We know $EA^2 = AF^2 + EF^2$

= 49 + 1 = 50

∴ EA = 5√2 km.

Q31

decrease stress level and reduce weight

Solution

The verb should appear first, so reduce should be before weight.

Since the sentence talks about clubs, reduces is wrong, so the answer is 2.

Q32

have succeeded in getting people who might have once thought the cosmetics were out of their league to buy and lease for some snazzy getups

Solution

Here, the subject, French luxury brands is plural in nature and hence the verb has to be plural.

Thus, 'have succeeded' is right.

It is idiomatic to say ***"in getting x... to do y"***. The correct construction should be ***"in getting people... to buy and use"***.

Hence, "have succeeded in getting people who might have once thought the cosmetics were out of their league to buy and use for some snazzy getups" is right.

Q33

short-lived

Solution

'Transient' means for a short term of period. The word that has the closest meaning would be short-lived.

Q34

Serious

Solution

Farcical: relating to or resembling farce, especially because of absurd or ridiculous aspects.

Absurd, Ludicrous and Facetious are synonyms of Farcical.

Q35

skeptic

Solution

The word agnostic means disbeliever or doubter. Theist refers to a person who has belief in god, dogmatist refers to extremist and conformist refers to follower. So the best answer is skeptic which means disbeliever or doubter.

Q36

Has the test been finished by you?

Solution

Solution: The given sentence is in active voice and present perfect tense. When the voice is changed, **you** becomes the object and **test** becomes subject. **Have** becomes **have been**. Since **test** takes **has**, the correct form would be **has been**.

The right answer is "Has the test been finished by you?"

Q37

Her lessons have been learned by her.

Solution

Solution:

The given sentence is in active voice and in present perfect tense.When the voice s changed, has learned becomes has been learned.

Lessons are in plural so the right form in passive is have been .The right answer is "**Her lessons have been learned by her**"

Q38

a

Solution

precious has consonant sound. The article used is 'a'

Q39

for

Solution

For - specify the use of something

The right answer is for

Q40

all states face similar industrial waste problems, the predominant industries and the regulatory environment of each state obviously determine

Solution

"all states face similar industrial waste problems, the predominant industries and the regulatory environment of each state obviously determine" is Correct This sentence makes the clear distinction between the problem all states share and the conditions each state faces; subject and verb agree.

"each state faces a similar industrial waste problem, their predominant industries and regulatory environment obviously determine"- Each state must be compared to all other states, not the states.

"all states face a similar industrial waste problem; their predominating industries and regulatory environment obviously determines"- Using a semicolon results in a sentence fragment; subject and verb do not agree

"each state faces similar industrial waste problems, the predominant industries and the regulatory environment of each state obviously determines"- Each state must be compared to all other states; subject and verb do not agree.

Q41

None of these

Solution

$200000x(1.08)^3 = 251942.4$

Q42

Rs.912

Solution

Whenever it is not mentioned whether we have to assume SI or CI we should assume SI.

For any amount, interest for the 1st three years @ 6% SI will be equal to 6 × 3 = 18%

Again, interest for next 4 years will be equal to 7 × 4 = 28%.

And interest for next 4 years (till 11 years) will be equal to 7.5 × 4 = 30%

So, total interest = 18 + 28 + 30 = 76%

So, total interest earned by him = 76% of the amount

= Rs.912

This calculation can be done very conveniently using the percentage rule as 75% + 1% = 900 + 12 = 912.

Q43

140

Solution

Number of squares (excluding 1*1 tile) is = 7*7 + 6*6 + 5*5 + ... + 1*1

= 7*8*15 / 6 = 140

Q44

$^9C_4 \times 6! \times 4$

Solution

The total no. of player in the football team is 11

The Captain & Vice -captain are already there in the out of 6 players, remaining 4 players can select form 9 player in the team so the total possibilities is $9C_4$

Six player can arrange themselves in 6! ways.

The manager can choose the photographer in 4 ways.

Hence the total possibilities is $9C_4*6!*4$.

Q45

22

Solution

minimum sum of 3 numbers= 24

maximum sum =129

now the series 24,29,34,.....,129

no of terms ={(129-24)/5}+1=22

Q46. Lenovo - S

Solution

7	R	Dell
6	P	Sony
5	S	Lenovo
4	L	Apple
3	D	Samsung
2	N	Acer
1	M	HP

Q47. Dell

Solution

7	R	Dell
6	P	Sony
5	S	Lenovo
4	L	Apple
3	D	Samsung
2	N	Acer
1	M	HP

Q48. The one who likes Apple

Solution

7	R	Dell
6	P	Sony
5	S	Lenovo
4	L	Apple
3	D	Samsung
2	N	Acer
1	M	HP

Q49. Only two people live between D and N

Solution

7	R	Dell
6	P	Sony
5	S	Lenovo
4	L	Apple
3	D	Samsung
2	N	Acer
1	M	HP

Q50. None

Solution

7	R	Dell
6	P	Sony
5	S	Lenovo
4	L	Apple
3	D	Samsung
2	N	Acer
1	M	HP

Q51. Also reflected subsequent environmental changes.

Solution

No Solution

Q52. 1 and 3 only

Solution

o Solution

Q53. The methods for determining past climatic conditions and then describes two of the earliest known methods.

Solution

No Solution

Q54. Data provided by dating marine sedimentation is more consistent with researchers' findings in other disciplines than is data provided by dating continental sedimentation.

Solution

No Solution

Q55. The high lake levels during the last ice age may have been a result of less evaporation rather than more precipitation.

Solution

No Solution

Q56. famously

Solution

first planned city of India a positive comment and the word famous is the right option

Q57. opportunities

Solution

The first blank should clearly have a 'positive' word .

Also the first blank should have a noun.

Hence, Opportunities is the correct answer

Q58. commuters

Solution

No Solution

Q59. affects

lution

ıss of life and property eventually affects the economy

Q60. suffer

Solution

No Solution

Section 2 - Pseudocode and Computer Fundamentals

Q1 32

Solution

First if block gets executed and update b,c as 16,16. So 0+16+16=32.

Q2 57

Solution

Inside firs loop:

When r=4 p and q updated to15,17

When r=5 p and q updated to 27 28

When r=6 p and q updated to 40 48 and loop terminates.

In second for loop:

When r=2 q updated to 15

When r=3 q updated to 16

When r=4 q updated to 17and loop terminates. Then 40+17=57

Q3 1

Solution

Here if condition gets failed so m^2 => 1 gets printed.

Q4 245

Solution

When fun1(23,222) inside while loop:

Iteration 1: x=201, y=44

Iteration 2: x=229,y=16

Iteration 3:x=245,y= 0 (y becomes 0 so loop stops)

So prints 245.

Q5

20

Solution

Initially p=2,q=6,r=5 after performing sequential operations values will be p=8,q= 8,r= 4 so 20 gets printed.

Q6

17

Solution

Initially p = 0, q = 3, r = 14.If condition gets failed so 0+3+14=>17.

Q7

54

Solution

When r=4 , p=13,q=21

When r=5 , p=23,q=31

Finally 23+31=>54 gets printed

Q8

0

Solution

No Solution

Q9

-2

Solution

Here m=1+4+3-1-9= -2 gets printed.

Q10

28

Solution

Inside loop:

In iteration 1: d=7+1+12=20, a=7+2=9, b=1+2=3

In iteration 2: d=9+3+12 = 24, a=9+2=11, b=3+2=5 same process repeated until condition gets failed.

In last iteration a=13 , b=7, d=28.

Q11

19

Solution

If block condition gets failed so 6+4+9=> 19.

Q12

true true true true false

22

Solution

No Solution

Q13

58

Solution

Initially a=8,b=3,c=5.

a=(3+5)&5 = 0

a=(0+6)+5 = 11

a=2+11=13

a=(3+5)+13 = 21

b=(3+8)+21=32

So 21+32+5=58.

Q14

3 3

Solution

Initially x = 11 , y = 10 then z and y updated with 3. I=3 because both x and z are non zero. So output is 3 3.

Q15

43

Solution

Initially p=2,q=8,r=7, p=8+1+8=17. Then if block gets executed and update r=9,p=26. So 26+8+9=43.

Q16

Uninterrupted Power Suplly

Solution

No Solution

Q17

Control unit, arithmetic logic unit, and primary storage

Solution

No Solution

Q18

program

Solution

General purpose computers are those that can be adapted to countless uses simply by changing their program. A general-purpose computer is a computer that is designed to be able to carry out many different tasks.

Q19 Crackers

Solution

No Solution

Q20 In RSA, both key are private

Solution

RSA is an asymmetric key protocol having one public key and one Private key.

Q21 UDP, TCP, UDP and TCP

Solution

No Solution

Q22 The physical layout of network devices

Solution

A network topology refers to the physical layout of network devices.

Q23 RIP (Routing Information Protocol)

Solution

No Solution

Q24 Link Establishment and Management

Solution

No Solution

Q25 Selective Repeat

Solution

No Solution

Q26 Destination or receiver synchronization loss

Solution

No Solution

Q27

81.3

Solution

From the Jacobson's algorithm,
Next Deviation $D_N = |\ IRTT - NRTT\ | = |45 - 60\ |\ sec = 15\ sec$
Expected deviation $DE = \alpha DI + (1 - \alpha)\ D_N$
$= 0.9*8 + 0.1*15$
$= 8.7$
$= 0.9 * 45 + (1 - 0.9)*60$
$= 46.5$
Time out $TO = ERTT + 4 * D_E = 46.5 + 4 \times 8.7 = 81.3$

Q28

1 MBPS

Solution

Given Slow start congestion protocol, so size of the sender window starts from 1MSS and increases exponentially.
So, for the 1st transmission, 1MSS = 1*1460
2nd transmission, 2MSS = 2*1460
3rd transmission, 4MSS = 4*1460
4th transmission, 8MSS = 8*1460
.
.
.
N^{th} transmission, N MSS = N*1460
Sum of all the data sent in N transmissions should be equal to
1000KB 1460(1 + 2 + 4 + 8 +....+ N) bytes = 1000KB
1 + 2 + 4 + 8 +.... + N = 1024000B/1460B
1 + 2 + 4 + 8 + ... + N = 701.369
It is in Geometric progression so sum of N terms in G.P is
$1(2^N - 1)/(2 - 1) = 701.369$
$2^N - 1 = 701.369$
$2^N = 702$ (approx) $\Rightarrow N = Ceil(\log_2 702) = 9.45$
N = 10
So we need to transmit 10 times to send all the 1000KB data file.
Therefore we need 10RTTs time
10*100ms = 1000KB
1sec = 1000KB/1000ms = 1MBPS

Q29

10/11

Solution

Given B = 40Mbps
L = 4000 bytes

$$T_{trans} = L/B = \frac{4000 \times 8\ bits}{(40*10^6\ bits/sec.)} = 800\ \mu sec$$

$$Efficiency = \frac{Trans}{Trans + Polling\ Time} \quad \frac{800\ \mu sec}{(800+80)\ \mu sec} = 10/11$$

Q30

33,555 seconds

Solution

Initial sequence number is 1 and it is incremented by 256000 every 2 sec,

We know that in TCP sequence no field is 32 bit long so total 232 sequence no are possible

now come 2 the question

256000 sequence no are consumed in 2 sec

so 232 sequence no can be consume in 232 *2 /256000 =33554.432 =33555 sec requires to wrap around

Section 3 - Coding

Q1

Test Case

Input

```
pineapplepen
3
pineapple pen applepen
```

Output

```
true
```

Weightage - 10

Input

```
abcde
3
ab bc ce
```

Output

```
false
```

Weightage - 10

Input

```
leetcodetech
4
leetcode tech leet code
```

Output

```
true
```

Weightage - 15

Input

```
interviewquestions
5
interview questions view terview inter
```

Output

```
true
```

Weightage - 25

Input

```
programmingcontest
2
programming contestprogwq
```

Output

```
false
```

Weightage - 25

Input

```
computer
4
compute puter com put
```

Output

```
true
```

Weightage - 15

Sample Input

Sample Output

applepenapple 2 apple pen	true

Sample Input **Sample Output**

catsandog 5 cats dog sand and cat	false

Solution

```
#include <stdio.h>
#include <stdlib.h>
#include <string.h>
#include <stdbool.h>

// Function to find the maximum length of strings in the word dictionary
int maxLength(char **wordDict, int dictSize) {
    int max_len = 0;
    for (int i = 0; i < dictSize; ++i) {
        int len = strlen(wordDict[i]);
        if (len > max_len) {
            max_len = len;
        }
    }
    return max_len;
}

// Function to check if the word can be segmented using the word dictionary
bool wordBreak(char *s, char **wordDict, int dictSize) {
    int n = strlen(s);
    bool *dp = (bool *)malloc((n + 1) * sizeof(bool));
    dp[0] = true;

    int max_len = maxLength(wordDict, dictSize);

    for (int i = 1; i <= n; i++) {
        dp[i] = false;
        for (int j = i - 1; j >= (i - max_len) && j >= 0; j--) {
            bool found = false;
            for (int k = 0; k < dictSize; ++k) {
                if (dp[j] && strncmp(wordDict[k], s + j, i - j) == 0 && strlen(wordDict[k]) == (i - j)) {
                    found = true;
                    break;
                }
            }
            if (found) {
                dp[i] = true;
                break;
            }
        }
    }

    bool result = dp[n];
    free(dp);
    return result;
}

int main() {
    char s[1001];
    scanf("%1000s", s);
```

```
    int dictSize;
    scanf("%d", &dictSize);

    char **wordDict = (char **)malloc(dictSize * sizeof(char *));
    for (int i = 0; i < dictSize; ++i) {
        wordDict[i] = (char *)malloc(101 * sizeof(char));
        scanf("%100s", wordDict[i]);
    }

    if (wordBreak(s, wordDict, dictSize)) {
        printf("true\n");
    } else {
        printf("false\n");
    }

    for (int i = 0; i < dictSize; ++i) {
        free(wordDict[i]);
    }
    free(wordDict);

    return 0;
}
```

```
#include <iostream>
#include <vector>
#include <string>
#include <algorithm>
using namespace std;

class Solution {
public:
    bool wordBreak(string s, vector<string>& wordDict) {
        int n = s.size();
        vector<bool> dp(n + 1, false);
        dp[0] = true;
        int max_len = 0;
        for (int i = 0; i < wordDict.size(); ++i) {
            max_len = max(max_len, static_cast<int>(wordDict[i].size()));
        }

        for (int i = 1; i <= n; i++) {
            for (int j = i - 1; j >= max(i - max_len - 1, 0); j--) {
                bool found = false;
                for (int k = 0; k < wordDict.size(); ++k) {
                    if (dp[j] && wordDict[k] == s.substr(j, i - j)) {
                        found = true;
                        break;
                    }
                }
                if (found) {
                    dp[i] = true;
                    break;
                }
            }
        }

        return dp[n];
    }
};

int main() {
    Solution sol;
    string s;
    cin >> s;
```

```
    int dictSize;
    cin >> dictSize;

    vector<string> wordDict(dictSize);
    for (int i = 0; i < dictSize; ++i) {
        cin >> wordDict[i];
    }

    if (sol.wordBreak(s, wordDict)) {
        cout << "true" << endl;
    } else {
        cout << "false" << endl;
    }

    return 0;
}
```

Q2

Test Case

Input

```
10
9 8 7 6 5 4 3 2 1 10
```

Output

```
9 8 7 6 5 4 3 2 1 10
false
```

Weightage - 25

Input

```
9
100 87 65 44 23 62 18 34 10
```

Output

```
100 87 65 44 23 62 18 34 10
false
```

Weightage - 25

Input

```
5
4 5 6 7 8
```

Output

```
4 5 6 7 8
true
```

Weightage - 10

Input

```
7
98 76 80 81 85 93 99
```

Output

```
98 76 80 81 85 93 99
true
```

Weightage - 15

Input

```
6
10 4 5 6 7 8
```

Output

```
10 4 5 6 7 8
true
```

Weightage - 15

Input **Output**

```
4
4 2 1 3
```

```
4 2 1 3
false
```

Weightage - 10

Sample Input **Sample Output**

```
3
4 2 3
```

```
4 2 3
true
```

Sample Input **Sample Output**

```
3
4 2 1
```

```
4 2 1
false
```

Solution

```
import java.util.Scanner;

class Solution {
    public static boolean checkPossibility(int[] nums) {
        int changeCount = 0;
        for(int index = 1; index < nums.length && changeCount <=1 ; index++){
            if(nums[index-1] > nums[index]){
                changeCount++;
                if(index-2<0 || nums[index-2] <= nums[index ])
                   nums[index-1] = nums[index];
                else nums[index] = nums[index-1];
            }
        }
        return changeCount<=1;
    }

    public static void main(String[] args) {
        Scanner scanner = new Scanner(System.in);

        int n = scanner.nextInt();

        int[] nums = new int[n];
        for (int i = 0; i < n; i++) {
            nums[i] = scanner.nextInt();
        }

        for (int i = 0; i < n; i++) {
            System.out.print(nums[i] + " ");
        }
        System.out.println();

        boolean result = checkPossibility(nums);
        System.out.print(result);

        scanner.close();
    }
}
```

```
def cb(arr):
    result=" ".join(map(str, arr))+" "
    print(result)
    c = 0
    for i in range(1, len(arr)):
        if arr[i] < arr[i - 1]:
            c += 1
        if c > 1:
            print("false")
            print()
            return

        if i == 1 or arr[i] >= arr[i - 2]:
            arr[i - 1] = arr[i]
        else:
            arr[i] = arr[i - 1]

    print("true")
    print()

n = int(input().strip())
arr = list(map(int, input().strip().split()))
cb(arr)
```